GOOGLE | 1POINT3ACRES | TOPIC 7 OF 10 | COMMON TOPIC

Strings, Parsing and Recursion

4 questions to study here, each on its own page, with the full interview answer and code

12 more fit this topic but are covered in another topic: they are listed as SKIP at the end of the contents

1	3
HARD	MEDIUM

How to use

- Questions are Hard first, then Medium, then Easy, newest last-asked date first.
- PDFs run from the one with the most questions to the fewest. A question has pages in only one PDF; the others list it as SKIP
- Every page has < Prev | Contents | Next > at the bottom. Tap a title in the Contents to jump to it.
- Select code and press Cmd/Ctrl+C to copy it.

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SKIP: these fit this topic but have their pages in another topic (12)

H	SKIP	Necklace Cut into Two Halves with Equal D/R Counts	topic 1: Arrays, Hashing and Two Pointers
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M	SKIP	Fence Painting Minimum Operations (Gemini Team)	topic 2: Dynamic Programming

STUDY HERE: topic 7, Strings, Parsing and Recursion

Also fits: Grids and Simulation (listed there as SKIP, no pages). Site tags: grid, backtracking, hard

HARD | 1Point3Acres | onsite-coding | 1 of 4

Matrix Flower Placement with House Adjacency

ASKED 2026-05-31 | 142 DAYS AGO | LAST 6 MONTHS

STUDY: TALK | NO LEETCODE FREQUENCY | PRIORITY 190/228

ABOUT THIS QUESTION

Topic: Strings, Parsing and Recursion

Optimal answer from: written and tested here for this exact question

Frequency: Low | Last asked: 2026-05-31 | Roles: SWE

Original: https://www.1point3acres.com/interview/problems/company/google/matrix-flower-placement-with-house-adjacency

QUESTION

EXAMPLES

```
H 0 0 0
H 0 0 0
0 0 0 H
0 0 0 0
```

The task is to fill some 0 cells with F while satisfying both the row/column constraint and each house's adjacency requirement.

APPROACH

Practice N-Queens with row/column sets and a recursive placement order.
Drill grid-neighbor helpers and constraint pruning before writing the full DFS.
Prepare the boolean version first, then generalize to collect all solutions.

INTERVIEW-DAY PLAN

CLARIFY FIRST

Ask whether each row and column holds at most one or exactly one flower, whether a house needs at least one or exactly one adjacent flower, whether adjacency is four-way, and whether to return one solution, a boolean, or all.

BRUTE FORCE TO SAY OUT LOUD

Try every subset of empty cells. Exponential.

OPTIMAL IDEA

Row-by-row backtracking like N-Queens: place at most one flower per row, track used columns, and check each house as soon as the row below it is decided (its neighbors are in rows h - 1 to h + 1). Prune at the first failed house. The boolean version stops at the first solution; the all-solutions version continues.

DRY-RUN THESE CASES

No houses, a house with no empty neighbor (impossible), two houses sharing one neighbor, a 1 x 1 grid.

LIKELY FOLLOW-UPS

Exactly-one rule, all solutions, or a faster exact-cover search.

PITFALLS

Checking a house before the next row is decided. Not restoring the cell on backtrack.

HOW TO THINK ABOUT IT

At most one flower per row and column is the N-Queens pattern: decide row by row which column (if any) gets the flower and remember the used columns. The extra rule about houses is local, so check a house as soon as the last row that can affect it is decided, and prune the branch if it fails.

Basis: N-Queens style backtracking (LeetCode 51/52) with extra local constraints. Assumptions stated explicitly: every house must have at least one flower in a 4-neighbour cell; rows and columns hold at most one flower (or exactly one if requested).

BRUTE FORCE

BRUTE FORCE · PSEUDO-CODE

```
def brute(grid):
    empty = cells_with(grid, 0)
    count = 0
    for k in range(len(empty) + 1):
        for flowers in combinations(empty, k):
            if not one_per_row_and_col(flowers):
                continue
            if houses_covered(grid, flowers):
                count += 1
    return count
```

TIME	SPACE
O(2^cells)	O(cells)

WHY IT IS SLOW

2^cells subsets: exponential. Backtracking row by row with a used-columns set and early house checks prunes almost all of them (still exponential in the worst case).

OPTIMAL SOLUTION · Row-by-row backtracking with used-column set and delayed house checks

TIME	SPACE
exponential worst case, h... exponential worst case, heavily pruned	O(rows + cols)

Code: see FULL INTERVIEW CODE below (the optimal solution is its main function).

FOLLOW-UP 1 What changes between 'at most one' and 'exactly one' flower per row and column?

With 'exactly one' the number of flowers is forced (rows must equal columns) and a row cannot be skipped, which prunes much harder, like N-Queens without the diagonal rule. With 'at most one' you also branch on skipping a row. State which reading you are using before writing code.

FOLLOW-UP 2 How would you speed up the search?

Prune earlier by checking each house as soon as its last neighbour row is decided (done above), order houses by how few candidate cells they have (most-constrained first), and keep bitmasks for used columns. For large grids this becomes a constraint satisfaction or exact-cover problem, so a SAT solver or DLX is the next step. **Cost:** the worst case stays exponential; most-constrained-first ordering and early house checks cut the explored states by orders of magnitude on typical grids.

FOLLOW-UP 3 Exactly one flower per row and column.

Add a flag that forbids skipping a row and requires as many columns as rows (otherwise no solution). The pruning is stronger, since every row must find a column.

FOLLOW-UP 4 Count the solutions instead of listing them.

Keep the same recursion but add one to a counter at the leaf instead of copying the grid; with a bitmask of used columns and memoisation on (row, mask, pending-house state) it can become a DP.

FULL INTERVIEW CODE · base question + every follow-up

One program for the whole question: the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```python
import itertools


# BRUTE FORCE (say it first): Choose any subset of empty
# cells and check all the rules | time O(2^cells)
def count_placements_brute(grid, exactly_one=False):
    R, C = len(grid), len(grid[0])
    empties = [
        (r, c)
        for r in range(R)
        for c in range(C)
        if grid[r][c] == "0"
    ]
    total = 0
    for k in range(len(empties) + 1):
        for chosen in itertools.combinations(
            empties, k
        ):
            rows, cols = [r for r, _ in chosen], [
                c for _, c in chosen
            ]
            if len(set(rows)) != len(rows) or len(
                set(cols)
            ) != len(cols):
                continue
            if exactly_one and not (
                len(rows) == R == C
            ):
                continue
            fl = set(chosen)
            if all(
                any(
                    (r + dr, c + dc) in fl
                    for dr, dc in (
                        (1, 0),
                        (-1, 0),
                        (0, 1),
                        (0, -1),
                    )
                )
                for r in range(R)
                for c in range(C)
                if grid[r][c] == "H"
            ):
                total += 1
    return total


# HELPERS for the base question
STEPS = ((1, 0), (-1, 0), (0, 1), (0, -1))


# BASE QUESTION (optimal): Row-by-row backtracking with
# used-column set and delayed ...
def place_flowers(grid, exactly_one=False,
                  find_all=False):
    # 'H' house, '0' empty. At most one F per row and
    # column; every house needs an F on a side.
    R, C = len(grid), len(grid[0])
    cells = [list(row) for row in grid]
    houses = [(r, c) for r in range(R) for c in range(C)
              if cells[r][c] == "H"]
    found = []

    def house_ok(r, c):
        # is some neighbour of this house a flower?
        for dr, dc in STEPS:
            a, b = r + dr, c + dc
            if 0 <= a < R and 0 <= b < C \
                    and cells[a][b] == "F":
                return True
        return False

    def row_houses_ok(r):
        return all(house_ok(hr, hc)
                   for hr, hc in houses if hr == r)

    def solve(r, used):
        # FOLLOW-UP 4: find_all=True lists every solution
        if found and not find_all:
            return
        # rows up to r-1 are decided, so row r-2 is final
        if r >= 2 and not row_houses_ok(r - 2):
            return
        if r == R:
            # FOLLOW-UP 3: exactly_one=True: a flower in
            # every column
            full = len(used) == C
            if row_houses_ok(R - 1) and (full or
                                         not exactly_one):
                found.append(["".join(x) for x in cells])
            return
        # try a flower in every free column of this row
```

```
        for c in range(C):
            if c not in used and cells[r][c] == "0":
                cells[r][c] = "F"
                solve(r + 1, used | {c})
                cells[r][c] = "0"
        # or leave the row empty (not allowed for exactly
        # one)
        if not exactly_one:
            solve(r + 1, used)

    solve(0, frozenset())
    if find_all:
        return found
    return found[0] if found else None


# FOLLOW-UP 4: Count the solutions instead of listing
# them.
def count_solutions(grid, exactly_one=False):
    return len(place_flowers(grid, exactly_one, True))


if __name__ == "__main__":
    grid = ["H00", "0H0", "00H"]
    sol = place_flowers(grid)
    assert sol is not None
    # every house has a flower beside it
    for r in range(3):
        for c in range(3):
            if grid[r][c] == "H":
                near = []
                for a, b in STEPS:
                    x, y = r + a, c + b
                    if 0 <= x < 3 and 0 <= y < 3:
                        near.append(sol[x][y])
                assert "F" in near
    # one flower per row and column
    assert all(row.count("F") <= 1 for row in sol)
    for c in range(3):
        assert [row[c] for row in sol].count("F") <= 1
    # a lone house in a 1 x 1 grid cannot be served
    assert place_flowers(["H"]) is None
```

```
    assert count_solutions(["00", "00"], True) == 2
    assert place_flowers(["000", "000"], True) is None
```

NOTES FROM CANDIDATES

This is an N-Queens-style DFS / backtracking problem with extra local constraints around houses.
Track occupied rows and columns, then validate the four-neighbor requirement either incrementally or at the leaf.
The two variants differ on at most one vs exactly one flower per row/column; state the assumption before coding.

PREPARATION

Practice N-Queens with row/column sets and a recursive placement order.
Drill grid-neighbor helpers and constraint pruning before writing the full DFS.
Prepare the boolean version first, then generalize to collect all solutions.

STUDY HERE: topic 7, Strings, Parsing and Recursion

Also fits: Grids and Simulation (listed there as SKIP, no pages). Site tags: recursion, simulation, medium

MEDIUM | 1Point3Acres | phone-screen, onsite-coding | 2 of 4

Mahjong Winning Hand Detection (14 Tiles)

ASKED 2026-10-09 | 11 DAYS AGO | LAST 30 DAYS

STUDY: READ | NO LEETCODE FREQUENCY | PRIORITY 210/228

ABOUT THIS QUESTION

Topic: Strings, Parsing and Recursion

Optimal answer from: written and tested here for this exact question

Frequency: Low | Last asked: 2026-10-09 | Roles: SWE

Original: https://www.1point3acres.com/interview/problems/company/google/mahjong-winning-hand-check

QUESTION

Mountain View onsite coding round 1: given a hand of 14 tiles, decide whether it is a winning mahjong hand. The candidate must clarify the rule set (typically: 4 melds of three identical tiles or three consecutive ranks + 1 pair).

Input: array of 14 integers representing tile values (clarify suit / range during the interview).
A winning hand consists of:

One pair (two identical tiles), and
Four melds, each being either:

A triplet (3 identical tiles), or
A run (3 consecutive ranks in the same suit).

Minority variant: A newer version instead specifies one pair plus two melds while retaining a 14-tile input. Because that accounts for only eight tiles, clarify the intended meld count and how the remaining six tiles should be handled before coding.

Output: boolean.

APPROACH

Practice writing the backtracking decomposer for tiles / counters; the canonical "expression add operators" recursion trains the same backtracking skeleton.
Have a tile-frequency-map mental model — sorting is the alternative but harder to reason about for runs across suits.
Pre-plan for the inevitable rules clarification: pull out at least 3 questions before coding.

HOW TO THINK ABOUT IT

A winning hand is one pair plus four melds (a triplet of identical tiles or a run of three consecutive ranks in one suit). Try every tile as the pair, then repeatedly take the smallest remaining tile: it can only be covered by a triplet of itself or by a run that starts at it, so there are at most two choices at each step and the search is tiny for 14 tiles.

HOW TO THINK ABOUT IT

Basis: Written for this question; the pair-plus-melds check is the same backtracking as LeetCode 846, Hand of Straights, plus a triplet option.

BRUTE FORCE

BRUTE FORCE · PSEUDO-CODE

```
def is_win(tiles):
    for t in set(tiles):
        if tiles.count(t) >= 2:
            rest = remove(tiles, [t, t])
            if all_melds(rest):
                return True
    return False
def all_melds(tiles):
    if not tiles:
        return True
    first = min(tiles)
    for meld in (triple(first), run(first)):
        if contains(tiles, meld):
            if all_melds(remove(tiles, meld)):
                return True
    return False
```

TIME	SPACE
roughly C(12,3) * C(9,3) ... roughly C(12,3) * C(9,3) * C(6,3) per pair	**O(1)**

WHY IT IS SLOW

Tries every set of three tiles at every step. The smallest remaining tile can only be in a triplet of itself or a run starting at it, so at most two choices remain.

OPTIMAL SOLUTION · Try each pair, then greedy-smallest with two options

TIME	SPACE
O(1) for 14 tiles (about 2^4 branches per pair)	**O(distinct tiles)**

Code: see FULL INTERVIEW CODE below (the optimal solution is its main function).

FOLLOW-UP 1 A variant needs only two melds plus a pair (8 tiles of 14).

Ask what happens to the other six tiles before coding: if they are free, check whether any 8 tiles form a pair and two melds with the same search over subsets of the 14. The rule set is intentionally vague, so confirm melds, honours and flowers first.

FULL INTERVIEW CODE · base question + every follow-up

One program for the whole question: the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```
import itertools
from collections import Counter


# BRUTE FORCE (say it first): Choose any three tiles for
# every meld | time roughly C(12,3) * C(9,3) * C(6,3) per
# pair
def mahjong_brute(hand):
    def is_meld(t):
        t = sorted(t)
        same = t[0] == t[1] == t[2]
        run = (
            t[0] // 10 == t[1] // 10 == t[2] // 10 < 3
            and t[1] == t[0] + 1
            and t[2] == t[0] + 2
        )
        return same or run

    def melds(tiles):
        if not tiles:
            return True
        for idx in itertools.combinations(
            range(len(tiles)), 3
        ):
            if is_meld(
                [tiles[i] for i in idx]
            ) and melds(
                [
                    t
                    for i, t in enumerate(tiles)
                    if i not in idx
                ]
            ):
                return True
        return False

    for i, j in itertools.combinations(
        range(len(hand)), 2
    ):
        if hand[i] == hand[j] and melds(
            [
                t
                for k, t in enumerate(hand)
                if k not in (i, j)
            ]
        ):
            return True
    return False


# HELPERS for the base question
def is_run_start(tile, count):
    # numbered suits are 0..2 with ranks 1..9; the tile
    # id is suit * 10 + rank. Honours (30+) never run.
    ok = tile < 30 and tile % 10 <= 7
    nxt, nxt2 = count.get(tile + 1), count.get(tile + 2)
    return ok and nxt and nxt2


def melds_left(count, need):
    # can the tiles in `count` form exactly `need` melds?
    live = [t for t in sorted(count) if count[t]]
    tile = live[0] if live else None
    if tile is None:
        return need == 0
    if need == 0:
        return False
    # the smallest tile is either in a triplet ...
    if count[tile] >= 3:
        count[tile] -= 3
        ok = melds_left(count, need - 1)
        count[tile] += 3
        if ok:
            return True
    # ... or starts a run of three
    if is_run_start(tile, count):
        for t in (tile, tile + 1, tile + 2):
            count[t] -= 1
        ok = melds_left(count, need - 1)
        for t in (tile, tile + 1, tile + 2):
            count[t] += 1
        return ok
    return False


# BASE QUESTION (optimal): Try each pair, then greedy-
# smallest with two options
# FOLLOW-UP 1: A variant needs only two melds plus a pair
# (8 tiles of 14).
def is_winning_hand(hand, melds=4):
    # FOLLOW-UP 1: melds=2 handles the 8-tile variant
    # one pair plus `melds` melds uses every tile
    if len(hand) != 2 + 3 * melds:
        return False
    count = Counter(hand)
    # try every tile value as the pair
    for tile in list(count):
        if count[tile] >= 2:
            count[tile] -= 2
            ok = melds_left(count, melds)
            count[tile] += 2
            if ok:
                return True
    return False


if __name__ == "__main__":
    win = [1, 1, 1, 2, 3, 4, 11, 12, 13,
           21, 22, 23, 31, 31]
    assert is_winning_hand(win)
    # 2, 3, 5 is not a run
    lose = [1, 1, 1, 2, 3, 5, 11, 12, 13,
            21, 22, 23, 31, 31]
    assert not is_winning_hand(lose)
    # honours only make triplets or the pair
    honours = [30, 30, 31, 31, 31, 32, 32, 32, 33, 33, 33,
               34, 34, 34]
    assert is_winning_hand(honours)
    # the smaller variant: a pair and two melds (8 tiles)
    assert is_winning_hand([5, 5, 1, 2, 3, 7, 7, 7], 2)
    # wrong tile count is never a win
    assert not is_winning_hand(win[:13])
```

NOTES FROM CANDIDATES

Clarify the rules — the prompt is intentionally sparse. Confirm: are honor tiles (winds/dragons) included? Do flowers count? Multi-suit?

Greedy + backtracking is the standard approach:

Try every possible pair (tile, tile); remove the pair.
On the remaining 12 tiles, repeatedly take the smallest remaining tile and try to form a triplet of it or a run starting at it; backtrack on failure.

NOTES FROM CANDIDATES

For 14 tiles this runs in well under 1 ms; the interviewer cares more about the case analysis and clean recursion than micro-optimizations.
The same algorithm extends to special hands (seven pairs / ????) if the interviewer adds them as a follow-up.

PREPARATION

Practice writing the backtracking decomposer for tiles / counters; the canonical "expression add operators" recursion trains the same backtracking skeleton.
Have a tile-frequency-map mental model — sorting is the alternative but harder to reason about for runs across suits.
Pre-plan for the inevitable rules clarification: pull out at least 3 questions before coding.

STUDY HERE: topic 7, Strings, Parsing and Recursion

Site tags: string, refactoring, medium

MEDIUM | 1Point3Acres | tech-screen | 3 of 4

Snake Case to Camel Case with Repository-Wide Refactor

ASKED 2026-07-14 | 98 DAYS AGO | LAST 6 MONTHS

STUDY: READ **NO LEETCODE FREQUENCY** **PRIORITY 213/228**

ABOUT THIS QUESTION

Topic: Strings, Parsing and Recursion

Optimal answer from: written and tested here for this exact question

Frequency: Single report | Last asked: 2026-07-14 | Roles: SWE

Original: https://www.1point3acres.com/interview/problems/company/google/snake-case-repository-refactor

QUESTION

Convert a valid snake_case identifier to camelCase, then design a safe repository-wide migration for variables and function names without changing third-party symbols, public interfaces, or the tests' ability to detect regressions.

Implement a conversion from a valid snake_case identifier to camelCase.
The base input is guaranteed valid: no leading underscore, trailing underscore, or doubled separator.
Example: foo_bar -> fooBar.
Extend the discussion to a repository too large for manual review: replace snake-case variables and functions across the codebase.
Explain how to verify that the repository still works after the automated rewrite, including when test-package identifiers are rewritten too.
Preserve third-party functions and public interfaces; explain how the migration distinguishes symbols that may change from contracts that must remain stable.

EXAMPLES

```
foo_bar -> fooBar
```

APPROACH

Implement the identifier conversion and enumerate edge cases before expanding scope.
Rehearse a migration plan covering symbol-aware rewriting, build and test validation, staged rollout, compatibility shims, and rollback.

HOW TO THINK ABOUT IT

Converting foo_bar to fooBar is easy. The risk is in the migration: a plain text replace also changes strings, comments, third-party names and public interfaces. Work on tokens (or the syntax tree) instead, rename only symbols defined in your own code, keep an explicit allow-list of public names, and prove nothing broke by running the tests before and after.

Basis: Written for this question; the conversion is a one-liner and the real work is a safe, token-based migration (not plain text replace).

BRUTE FORCE

BRUTE FORCE · PSEUDO-CODE

```
for path in source_files(repo):
    text = read(path)
    for word in find_snake_words(text):
        if not is_third_party(word):
            new = to_camel(word)
            text = text.replace(word, new)
    write(path, text)
```

TIME	SPACE
O(size of the repository)	O(1)

WHY IT IS SLOW

Also rewrites strings, comments, third-party calls and public interface names. Renaming only the identifiers the code defines, found with the syntax tree and tokens, keeps those intact.

OPTIMAL SOLUTION · Convert names, rename only defined identifiers via tokens

TIME	SPACE
O(size of the source)	O(names)

Code: see FULL INTERVIEW CODE below (the optimal solution is its main function).

FOLLOW-UP 1 How do you prove the migration is safe?

Run the full test suite before and after (green both times), compare the public API surface (exported names) before and after, and migrate in small reviewable commits with a codemod that only touches owned symbols. Anything reachable by reflection, serialisation, config files or other repositories needs an explicit alias or a deprecation period rather than a silent rename.

FULL INTERVIEW CODE · base question + every follow-up

One program for the whole question: the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```python
import re
import ast
import io
import tokenize


# BRUTE FORCE (say it first): Plain text replace across
# the files | time O(size of the repository)
def snake_rewrite_brute(source):
    return re.sub(
        r"\b[a-z]+(?:_[a-z0-9]+)+\b",
        lambda m: snake_to_camel(m.group()),
        source,
    )


# BASE QUESTION (optimal): Convert names, rename only
# defined identifiers via tokens
def snake_to_camel(name):
    # foo_bar -> fooBar (input is a valid snake_case name)
    head, *rest = name.split("_")
    caps = [p[:1].upper() + p[1:] for p in rest]
    return head + "".join(caps)


# FOLLOW-UP 1: How do you prove the migration is safe?
def owned_names(source, keep=()):
    # names this code defines itself: safe to rename
    names = set()
    for node in ast.walk(ast.parse(source)):
        if isinstance(node, ast.FunctionDef):
            names.add(node.name)
            args = node.args.args + node.args.kwonlyargs
            names.update(a.arg for a in args)
        elif isinstance(node, ast.Name):
            if isinstance(node.ctx, ast.Store):
                names.add(node.id)
    # only real snake_case, never dunder or private names
    return {n for n in names if "_" in n.strip("_")
            and not n.startswith("_") and n not in keep}


# FOLLOW-UP 1: How do you prove the migration is safe?
def rename_identifiers(source, keep=()):
    owned = owned_names(source, keep)
```

```python
    out, prev = [], None
    # work on tokens: strings and comments are untouched
    for tok in tokenize.generate_tokens(
            io.StringIO(source).readline):
        # skip attribute names (after a dot): they may be
        # third-party, so they stay as written
        dot = prev is not None and prev.string == "."
        is_name = tok.type == tokenize.NAME
        if is_name and tok.string in owned and not dot:
            new = snake_to_camel(tok.string)
            tok = tok._replace(string=new)
        out.append(tok)
        prev = tok
    return tokenize.untokenize(out)


if __name__ == "__main__":
    assert snake_to_camel("foo_bar") == "fooBar"
    assert snake_to_camel("a_b_c") == "aBC"
    assert snake_to_camel("name") == "name"
    src = (
        "import os\n"
        "def load_data(file_path):\n"
        "    full_path = os.path.join(file_path, 'x_y')\n"
        "    return full_path  # load_data is fine\n"
        "result_value = load_data('a')\n")
    new = rename_identifiers(src, keep={"file_path"})
    assert "def loadData(file_path)" in new
    assert "fullPath = os.path.join" in new
    # strings, comments and third-party calls are intact
    assert "'x_y'" in new and "# load_data is fine" in new
    assert "resultValue = loadData('a')" in new
    # the migrated code still runs
    scope = {}
    exec(compile(new, "x", "exec"), scope)
    assert scope["resultValue"] == "a/x_y"
```

NOTES FROM CANDIDATES

The implementation is intentionally short. Most of the round is spent on scoping a safe refactor, validation strategy, symbol ownership, and compatibility boundaries.
Clarify whether names appear only as parsed identifiers or may also occur in strings, generated code, configuration, reflection, and serialized data.
A text replacement is not enough to distinguish local symbols from external APIs or public contracts.

PREPARATION

Implement the identifier conversion and enumerate edge cases before expanding scope.
Rehearse a migration plan covering symbol-aware rewriting, build and test validation, staged rollout, compatibility shims, and rollback.

STUDY HERE: topic 7, Strings, Parsing and Recursion

Also fits: Grids and Simulation (listed there as SKIP, no pages). Site tags: string, simulation, medium

MEDIUM | 1Point3Acres | onsite-coding | 4 of 4

Google Doc Line Wrap (Word Wrap with Whitespace and Newlines)

ASKED 2026-02-01 | 261 DAYS AGO | OLDER

STUDY: READ | NO LEETCODE FREQUENCY | PRIORITY 202/228

ABOUT THIS QUESTION

Topic: Strings, Parsing and Recursion

Optimal answer from: written and tested here for this exact question

Frequency: Single report | Last asked: 2026-02-01 | Roles: SWE

Original: https://www.1point3acres.com/interview/problems/company/google/document-line-wrap

QUESTION

EXAMPLES

```
text  = "Hello world\nFoo bar baz qux"
width = 10
Line 1: "Hello"             (5 chars; "world" + space
    would be 11 -> break)
Line 2: "world"
Line 3: "Foo bar"            (7 chars; "baz" -> "Foo
    bar baz" = 11 -> break)
Line 4: "baz qux"
Output: 4
```

APPROACH

Implement a clean tokenizer that treats \n as a special token and yields plain words otherwise.
Test cases: width = 1, text = "a\n\n\nb", text = " leading", text with words exactly width chars long.
Be ready to extend to "return the rendered lines" (string list), not just the count, as a likely follow-up.

INTERVIEW-DAY PLAN

CLARIFY FIRST

Ask whether the width counts characters in a monospace font, how to treat explicit newlines, multiple spaces and leading spaces, what to do with a word longer than the width, and whether to return the number of lines or the lines themselves.

BRUTE FORCE TO SAY OUT LOUD

Build the text line by line by trying to add each word and measuring. That is already the greedy algorithm.

OPTIMAL IDEA

Tokenize into words and newline tokens. Keep the current line length; for each word, if it fits (with a space unless the line is empty) add it, otherwise start a new line. A newline token always ends the current line. Count lines (empty lines count). Greedy is expected; mention Knuth's minimum-raggedness DP only if asked. O(n).

INTERVIEW-DAY PLAN

DRY-RUN THESE CASES

The example giving 4. width = 1. Consecutive newlines. A word exactly the width. A word longer than the width.

LIKELY FOLLOW-UPS

Return the lines, justify the text, or minimize raggedness.

PITFALLS

Dropping empty lines. Adding a trailing space at the end of a line.

HOW TO THINK ABOUT IT

Greedy is optimal for the question 'fewest lines': put as many words on a line as fit. The decision per word is only 'fits with a space', 'fits as the first word on the line', or 'start a new line', so track the current line length, not the line text. Treat newline as a hard break and decide up front what empty lines and long words do.

Basis: Written; greedy word wrap (LeetCode 68, Text Justification, without the justification step). Assumed rules (confirm them): monospace width in characters; words are separated by whitespace; '\n' always ends a line and an empty line counts; a word longer than the width gets its own line.

BRUTE FORCE

BRUTE FORCE · PSEUDO-CODE

```
def wrap(text, width):
    count = 0
    for paragraph in text.split("\n"):
        line = ""
        for word in paragraph.split():
            joined = len(line) + 1 + len(word)
            if line and joined > width:
                count += 1  # finish the line
                line = word
            else:
                line = line + " " + word
                line = line.strip()
        count += 1  # last line
    return count
```

TIME	SPACE
O(total characters * words per line)	O(width)

WHY IT IS SLOW

Re-measuring a joined string for every word costs extra copying. Keeping a running length makes the whole thing O(total characters).

OPTIMAL SOLUTION · Greedy with a running line length

TIME	SPACE
O(n)	O(1) extra beyond the output

Code: see FULL INTERVIEW CODE below (the optimal solution is its main function).

FOLLOW-UP 1 Return the lines, not only the count.

Keep the same loop but append the finished line to a list (the brute force above already does it). Build each line from a list of words and join once, so each word is copied once.

FOLLOW-UP 2 A word longer than the width must be broken.

Before placing a word, while it is longer than the width flush the current line and emit width-sized chunks of the word, then continue with the remainder as a normal word.

FOLLOW-UP 3 Minimise raggedness instead of lines.

That is Knuth's word-wrap DP: dp[i] is the best cost of laying out words i.. with cost (spaces left)^2 on each line except the last; O(n * words per line). Greedy is not optimal for this objective.

FULL INTERVIEW CODE · base question + every follow-up

One program for the whole question: the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```
# BRUTE FORCE (say it first): Build each line as a string
# and measure it | time O(total characters * words per
# line)
def wrap_lines_brute(text, width):
    lines = []
    for paragraph in text.split("\n"):
        words = paragraph.split()
        cur = ""
        for w in words:
            trial = (cur + " " + w) if cur else w
            if len(trial) <= width or not cur:
                cur = trial
            else:
                lines.append(cur)
                cur = w
        lines.append(cur)
    return lines


# HELPERS for the base question
# NEEDED BY THE BASE QUESTION: it is also the answer to
# the follow-up below
# FOLLOW-UP 1: Return the lines, not only the count.
def wrap_lines(text, width, break_long=False):
    lines = []
    # a newline always starts a new paragraph (and a new
    # line)
    for paragraph in text.split("\n"):
        cur = ""
        words = paragraph.split()
        if break_long:
            pieces = [split_long(w, width) for w in words]
            words = [p for chunk in pieces for p in chunk]
        for w in words:
            if not cur:
                cur = w
            elif len(cur) + 1 + len(w) <= width:
                # the word and one space still fit
                cur += " " + w
            else:
                lines.append(cur)
                cur = w
        # the last (or only, or empty) line of the
        # paragraph
        lines.append(cur)
    return lines


# NEEDED BY THE BASE QUESTION: it is also the answer to
# the follow-up below
# FOLLOW-UP 2: A word longer than the width must be
# broken.
def split_long(word, width):
    # a word longer than the line is cut into pieces
    starts = range(0, len(word), width)
    return [word[i:i + width] for i in starts]


# BASE QUESTION (optimal): Greedy with a running line
# length
def wrap_count(text, width):
    return len(wrap_lines(text, width))


# FOLLOW-UP 3: Minimise raggedness instead of lines.
def min_raggedness(words, width):
    # Knuth: minimise the squared spare space of every
    # line
    # except the last; greedy is NOT optimal for this goal
    n = len(words)
    INF = float("inf")
    # cost[i]: best total cost for words[i:]
    cost, nxt = [INF] * (n + 1), [0] * (n + 1)
    cost[n] = 0
    for i in range(n - 1, -1, -1):
        length = -1
        for j in range(i, n):
            length += len(words[j]) + 1
            if length > width:
                break
            spare = width - length
            # the last line costs nothing
            extra = 0 if j == n - 1 else spare ** 2
            if extra + cost[j + 1] < cost[i]:
                cost[i] = extra + cost[j + 1]
                nxt[i] = j + 1
    lines, i = [], 0
    while i < n:
        lines.append(" ".join(words[i:nxt[i]]))
        i = nxt[i]
    return cost[0], lines


if __name__ == "__main__":
    text = "Hello world\nFoo bar baz qux"
    assert wrap_lines(text, 10) == [
        "Hello", "world", "Foo bar", "baz qux"]
    assert wrap_count(text, 10) == 4
    # blank lines count as lines
    assert wrap_count("a\n\nb", 5) == 3
    # a long word is split when asked
    cut = wrap_lines("abcdefgh", 3, True)
    assert cut == ["abc", "def", "gh"]
    assert wrap_lines("abcdefgh", 3) == ["abcdefgh"]
    # the best layout can beat the greedy one
    cost, lines = min_raggedness(
        "aaa bb cc ddddd".split(), 6)
    assert lines == ["aaa", "bb cc", "ddddd"]
    assert cost == 9 + 1
```

NOTES FROM CANDIDATES

Single-pass tokenizer + accumulator: track curLen; on each token, decide "fits with leading space", "fits without space" (start of line), or "flush and start new line".
Watch out for \n right after another \n (empty line counts), \n at the end of input, leading whitespace, multiple consecutive spaces inside a line.
Don't try to handle font / kerning / hyphenation — the prompt assumes monospace width counted in characters.
Greedy packing is the expected answer (LC 68 style). Mention DP word-wrap (Knuth's algorithm) only if the interviewer pushes on 'minimum raggedness'; greedy is optimal-enough and what graders look for. Watch edge cases: a single word longer than W, an explicit newline forcing a break mid-pack, trailing whitespace handling.

PREPARATION

Implement a clean tokenizer that treats \n as a special token and yields plain words otherwise.
Test cases: width = 1, text = "a\n\n\nb", text = " leading", text with words exactly width chars long.
Be ready to extend to "return the rendered lines" (string list), not just the count, as a likely follow-up.